

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455851.647 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1761 +/- 0.0074
Transit depth (Rp/Rs)^2	3.1 +/- 0.26 [%]
Orbital Inclination (inc)	89.95 +/- 0.42
Airmass coefficient 1 (a1)	0.923 +/- 0.011
Airmass coefficient 2 (a2)	0.07 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	#1 - [321, 320]
Optimal Aperture	3.39
Optimal Annulus	5.92
Transit Duration (day)	0.09639 +/- 0.00074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1761 ± 0.0074 vs 0.1592 (deviation 2.28σ)
Duration fit vs published	0.09639 d vs 0.0946 d (deviation 2.44σ)
Mid-time O–C	1.7 min
Depth SNR	23.8
Residual scatter	1.25 %

Light curve

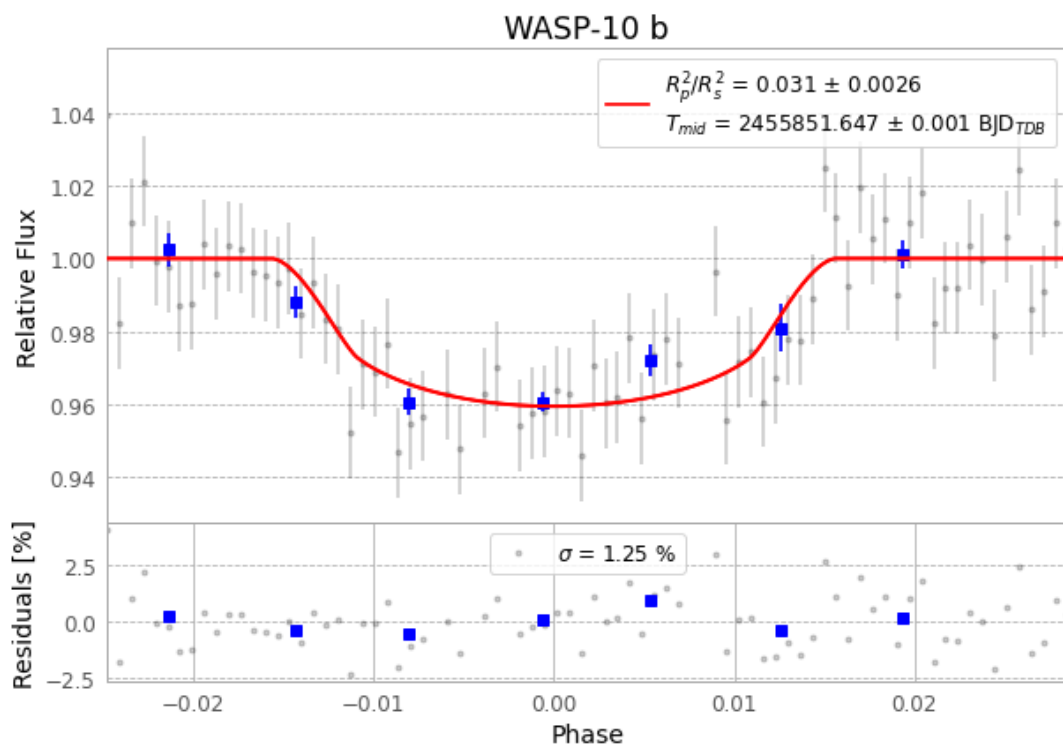


Figure 1 — FinalLightCurve_WASP-10 b_2011-10-16.png (SHA-256 d5a624ec4d67d894...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 196], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 faa385db7c813465...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

80 FITS frames (52 MB), WASP-10111017013313.FITS → WASP-10111017053012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-111017.log (f30c2d99ed18...), FinalLightCurve_WASP-10 b_2011-10-16.png (d5a624ec4d67...), FinalLightCurve_WASP-10 b_2011-10-16.csv (f75c16c3a0bc...)

Compute resources

Wall time 82.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:10Z.